

Point thèse February 3, 2026(MV)

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Multivariate case

- Revenir sur la logique générale utilisée.
- Choix du modèle paramétrique et effet sur les résultats.

Reminder on regular variations

- Let denote $\mathcal{R}_\tau$ the set of regularly varying functions v of index $\tau > 0$ verifying $\lim_{u \rightarrow +\infty} \frac{v(tu)}{v(u)} = t^\tau$ for every $t > 0$. $\mathcal{C}_0 = (0, +\infty]^2$

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- Let $M_0(\mathcal{C})$ be the set of Borel measures on $\mathcal{C}_0$.
- Following definition 3.1, a random element X in $\mathcal{C}$ is regularly varying if and only if there exists a function $W \in \mathcal{R}_\alpha$ and a nonzero measure $\omega \in M_0(\mathcal{C})$ such that:

$$W(t)^{-1} \mathbb{P}(Xt^{-1} \in \cdot) \rightarrow \omega(\cdot)$$

as $t \rightarrow +\infty$.

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as $t \rightarrow +\infty$.

- ω must be homogeneous with $\omega(tA) = t^{-\alpha} \omega(A)$

- We work in the bivariate case with $d = 2$ and $\ell = \|\cdot\|$, we apply a different polar decomposition $\mathbb{B}_0^2 \mapsto ([0, +\infty)^2 \setminus \{\mathbf{0}\}) \times \mathbb{S}^2$ given by $f_1, f_2 \mapsto (\|f_1\|, \|f_2\|), (\frac{f_1}{\|f_1\|}, \frac{f_2}{\|f_2\|})$.

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- Following (Kim and Kokoszka 2024), μ is decomposed into two measures: one for the law of

$$(\Theta_{M,1}, \Theta_{M,2}) = (\mathbf{Z}_{M,1}/\ell(\mathbf{Z}_{M,1}), \mathbf{Z}_{M,2}/\ell(\mathbf{Z}_{M,2}))$$

denoted μ_Θ and another for the law of

$$(\ell(\mathbf{Z}_{M,1})/b(t), \ell(\mathbf{Z}_{M,2})/b(t))$$

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denoted μ_R .

- μ_R is by definition homogeneous of index α and $b \in \mathcal{R}_{\alpha-1}$.

- **Extreme events:** Define a function $g : \mathbb{R}_+^2 \mapsto \mathbb{R}$ such that $g_\ell(\mathbf{Z}_M) = g(\ell(\mathbf{Z}_{M,1}), \ell(\mathbf{Z}_{M,2}))$ defines a norm for the space $\mathbb{B}_0^2$.

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- We obtain the conditional independence between the angular and radial components:

$$\left(\frac{\mathbf{Z}_{M,1}}{\ell(\mathbf{Z}_{M,1})}, \frac{\mathbf{Z}_{M,2}}{\ell(\mathbf{Z}_{M,2})} \right) \perp\!\!\!\perp (\ell(\mathbf{Z}_{M,1}), \ell(\mathbf{Z}_{M,2})) \quad (1)$$

knowing $g_\ell(\mathbf{Z}_M) > u_{\ell,g}$.

Polar decomposition

- We note $\Omega_M^g = (\Theta_{M,1}, \Theta_{M,2}) \mid g_\ell(\mathbf{Z}_M) > u_{\ell,g}$ the angles of extreme time series. By definition, we have $\Theta_{M,1}, \Theta_{M,2} \in \mathbb{R}_+^{37}$.

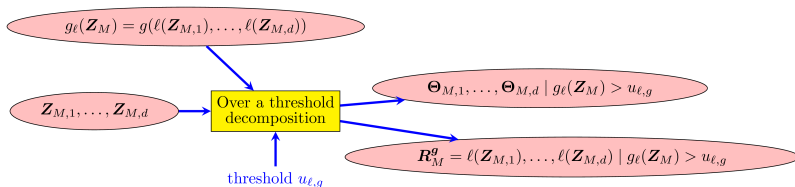


Figure 1: Construction of the polar coordinate representation from transformed data (corresponding sections indicated near the boxes)

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- We note $\mathbf{R}_M^g = (\ell(\mathbf{Z}_{M,1}), \ell(\mathbf{Z}_{M,2})) \mid g_\ell(\mathbf{Z}_M) > u_{\ell,g}$ the radial components defining the extreme aspect. We have $R_{M,1}^g, R_{M,2}^g \in \mathbb{R}_+$.

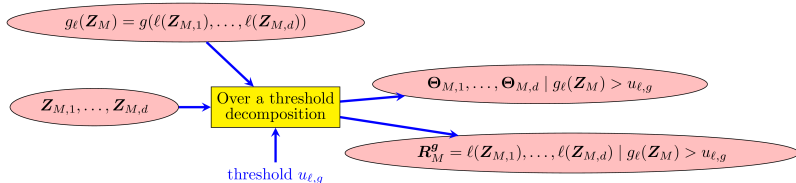


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- Following (Dombry and Ribatet 2015) and (Kim and Kokoszka 2024), one choice of $b(t)$ is the function verifying $\mathbb{P}(g(\mathbf{R}_M^g) > b(t)) = t^{-1}$.

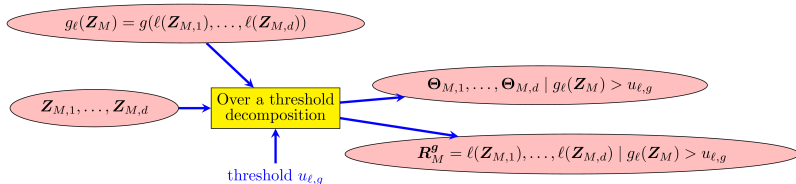


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Extremes of the radial components

- Following definition 3.1 from (Segers, Zhao, and Meinguet 2017), since ν is a homogenous measure defined on $\mathcal{C}_0$ of index 1, the vector $(\ell(Z_{M,1}), \ell(Z_{M,2}))$ is **regularly varying**.

Modulus definition

Let d be a distance. Following definition 2.2 from (Segers, Zhao, and Meinguet 2017), $g : \mathbb{R}_+^2 \mapsto \mathbb{R}_+$ is a **modulus** if:

- ① g is continuous
- ② g is homogeneous: $g(\lambda x) = \lambda g(x) \quad \forall \lambda > 0$.
- ③ $\forall \varepsilon > 0$, we have $\inf\{g(x)/d(x, \mathbf{0}) > \varepsilon\} > 0$.

Thus, we can use norms but we cannot use the minimum.

Regular variations and Pareto laws

Let $\theta(X) = Xg(X)^{-1}$. Following (Proposition 3.1) of (Segers, Zhao, and Meinguet 2017), we have an equivalence between:

- X is regularly varying of index α .
- $\mathcal{L}(g(X)/u, \theta(X) \mid g(X) > u) \xrightarrow{d} \text{Pareto}(\alpha) \otimes H$. **Asymptotic independence** between the radial and the angular components with H a probability measure.

Consequence

- It implies following (Proposition 3.1) of (Segers, Zhao, and Meinguet 2017) that for every modulus g ,

$$\left(\frac{g(\ell(\mathbf{Z}_{M,1}), \ell(\mathbf{Z}_{M,2}))}{u} \mid g(\ell(\mathbf{Z}_{M,1}), \ell(\mathbf{Z}_{M,2})) > u \right)$$

will also follow a Pareto(1)

Various cases of extremal dependence structure

- We define here a "scenario" as a situation where some coordinates of the vector are extremes ($> u_{\ell,g}$) whereas the rest take low values ($< u_{\ell,g}$).
- The number of possible scenarios depends on the extremal dependence (Table 1). For instance, if (X, Y) exhibits asymptotic independence, X cannot be extreme when Y is extreme, which gives two different scenarios. See document SimADAI.pdf for more details.

A.Dependencies	Complete AI	Partially AD	Complete AD
Model used	Gaussian	Mixture model	MEV model
Number of scenarios	d	$< d$	1

Table 1: The several raised cases with $d = 2$

Approach angular components

We denote

$$\bar{\Omega}_M^g = (\Omega_{M,1}^1, \dots, \Omega_{M,1}^{37}, \dots, \Omega_{M,d}^1, \dots, \Omega_{M,d}^{37})$$

We apply PCA on the standardized vector.

Application

Warning

- Reminder: Let X be a random variable following a Generalized Pareto distribution of parameters μ (location), $\sigma > 0$ (scale), γ (shape). Let $z = 1 + \gamma\left(\frac{x-\mu}{\sigma}\right)$, the cdf of X for $x > \mu$ and $z > 0$ is:

$$\mathbb{P}(X \leq x) = 1 - (z)^{-\frac{1}{\gamma}}$$

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- We get what we will call from now a Pareto margin when $\mu = 0$, $\sigma = 1$ and $\gamma = 1$.
- It is different from unit Pareto law since $\mu = 1$ for the latter.

Whitening

- Use the method with $\tilde{\mathbf{X}}_{M,1}$ representing the surge (S) time series, $\tilde{\mathbf{X}}_{M,2}$ representing significant wave height (Hs) series. We denote $\tilde{\mathbf{X}}_M^t$ the vector $(\tilde{X}_{M,1}^t, \tilde{X}_{M,2}^t) \in \mathbb{R}^2$.

Whitening

- Use the method with $\tilde{\mathbf{X}}_{M,1}$ representing the surge (S) time series, $\tilde{\mathbf{X}}_{M,2}$ representing significant wave height (Hs) series. We denote $\tilde{\mathbf{X}}_M^t$ the vector $(\tilde{X}_{M,1}^t, \tilde{X}_{M,2}^t) \in \mathbb{R}^2$.
- For each value of t , we model the temporal dependence within and between each variable, with respect to M , by an vector autoregressive model (VAR) of order p , leading to

$$\tilde{\mathbf{X}}_M^t = \sum_{j=1}^p \mathbf{A}_j^t \tilde{\mathbf{X}}_{M-\Delta j}^t + \epsilon_M^t \quad (2)$$

for $M = 1 + m\Delta$ with $m = 1, 2, \dots$ and $\mathbf{A}_j^t \in \mathcal{M}_{2 \times 2}(\mathbb{R})$.

Effect of autoregressive models

- Figure 2 shows that we remove the correlations for the surge at time $t = 19$ with our $VAR(1)$ model.

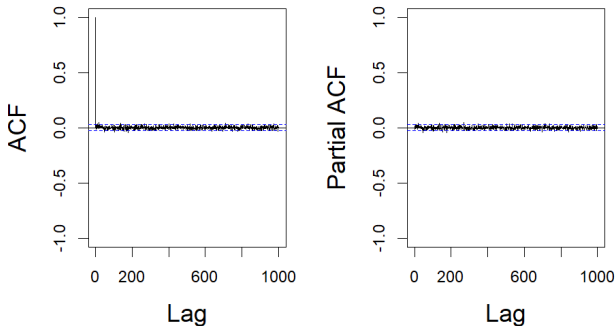


Figure 2: ACF and PACF for each event at time $t = 19$ with a lag in terms of tidal cycle index for the residuals of the surge

Effect of autoregressive models

- We can see the same type of result with wind speed on Fig. 3.

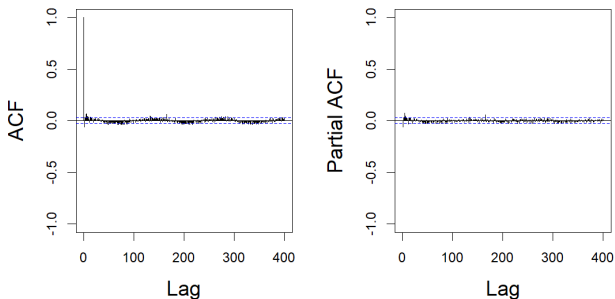


Figure 3: ACF and PACF for each event at time $t = 19$ with a lag in terms of tidal cycle index for the residuals of H_s

Cross-correlations

- Figure 4 presents the effect of the VAR model on cross-correlations.

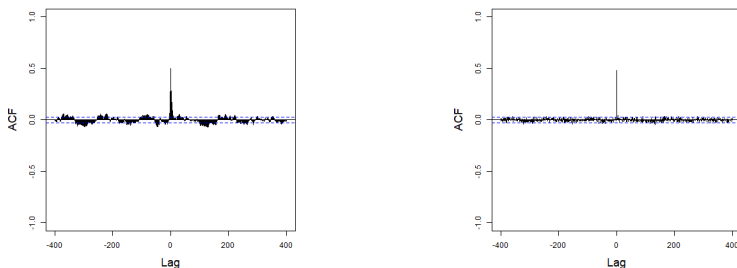


Figure 4: a) $(\tilde{X}_{M,1}^{13}, \tilde{X}_{M,2}^{19})$, b) $(\varepsilon_{M,1}^{13}, \varepsilon_{M,2}^{19})$

Cross-correlations

- Figure 5 presents the effect of the VAR model on the cross-correlations for the same forcing condition.

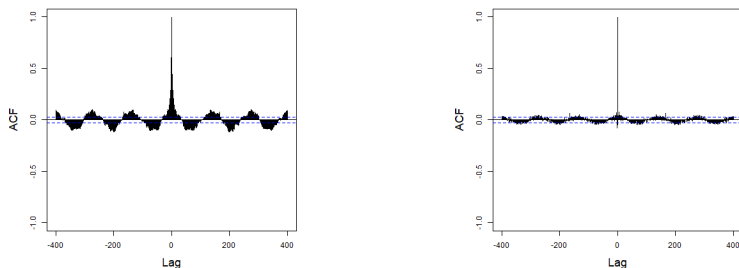


Figure 5: a) $(\tilde{X}_{M,1}^{13}, \tilde{X}_{M,1}^{19})$, b) $(\varepsilon_{M,1}^{13}, \varepsilon_{M,1}^{19})$

Marginal transformation

- Since we approach efficiently for each $t = 1, \dots, 37$ and $k = 1, 2$ the cdf of $\varepsilon_{t,k}$, we apply the transformation $\mathcal{T}_k$ for $k = 1, 2$. We get **Pareto** margins.

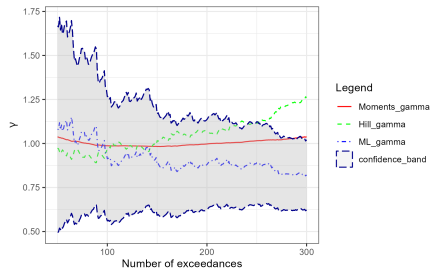
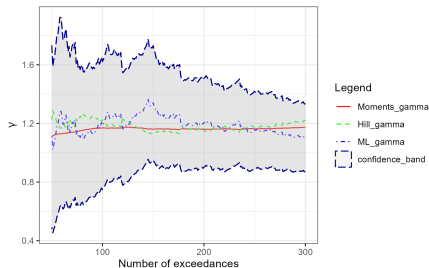


Figure 6: Estimation of γ for: $\ell(\mathbf{Z}_{M,1})$; $\ell(\mathbf{Z}_{M,2})$ (solid line: moments estimator, dotted line: Hill estimator, dashed line: MLE estimator)

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- Shift in the tail behavior, emphasized by the convergence of the Hill estimator (Fig. 6) of $\ell(\mathbf{Z}_{M,k})$ for each $k = 1, 2$.

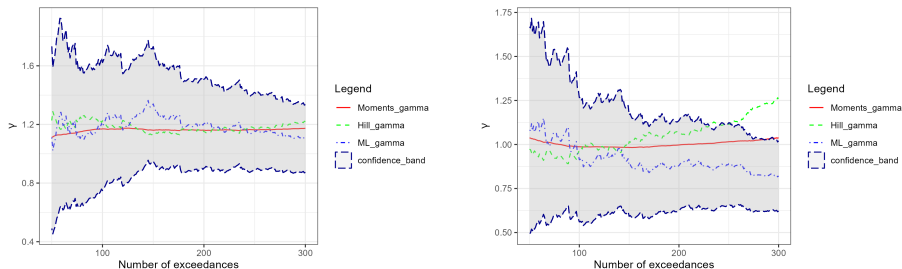


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Choice of extreme time series

- Choose the threshold $u_{\ell,g}$ that defines the set of extreme observations with $g_{\ell}(\mathbf{Z}_M) > u_{\ell,g}$. Convergence to a Pareto law.

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- Following (Danielsson et al. 2016), select the rank r minimising a metric comparing the cdf of extreme time series with a Pareto cdf, which gives us $r = 147$.

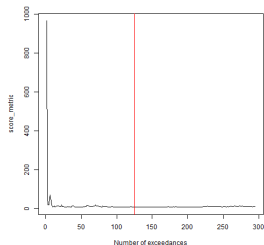


Figure 7: Evolution of the metric score between the cdf of $g_{\ell}(\mathbf{Z}_M)$ and a unit Pareto cdf

Polar decomposition

- Since we select extreme time series, we now compute the angular component of both Surge and Hs.



Figure 8: Sampling of 20 surge time series with decomposition: $\mathbf{Z}_{M,1}$ (left panel), $\Theta_{M,1}$ (right panel)

Polar decomposition

- Since we select extreme time series, we now compute the angular component of both Surge and Hs.
- Figure 8 shows the results for the surge.

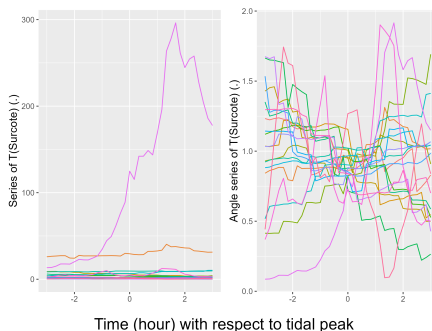


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polar decomposition (2)

Figure 9 shows the results for Hs.

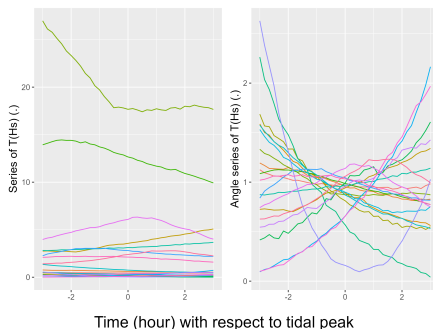


Figure 9: Sampling of 20 Hs time series with decomposition: $Z_{M,2}$ (left panel), $\Theta_{M,2}$ (right panel)

Marginal transformations of radial components

$(\ell(\mathbf{Z}_{M,1}), \ell(\mathbf{Z}_{M,2}))$

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- First, we take Pareto margins for $Z_{M,k}^t$ for $k = 1, 2$ and $t = 1, \dots, 37$.
- Then, we approach the law of $\log(\ell(\mathbf{Z}_{M,1}) + 1)$ and the one of $\log(\ell(\mathbf{Z}_{M,2}) + 1)$ with EGPD laws from (Leo R Belzile 2023). This implementation can only be applied to positive values.

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- Approaching the cdf $\log(\ell(\mathbf{Z}_{M,j}) + 1)$ gives much better results than $\ell(\mathbf{Z}_{M,j})$ for $j = 1, 2$.

EGPD fitting

- Figure 10 provides some fitting results of $\log(\ell(\mathbf{Z}_{M,1}))$ and $\log(\ell(\mathbf{Z}_{M,2}))$.
- The data points remain near the bisector.

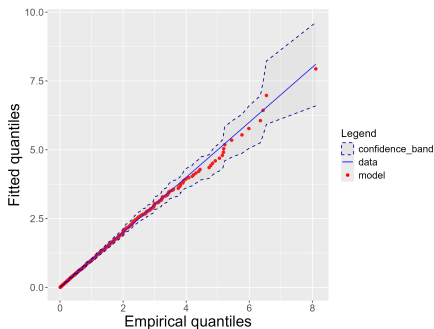
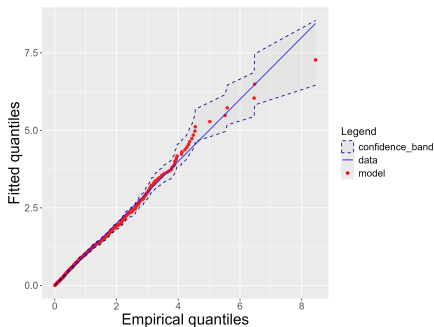


Figure 10: EGPD fitting: a)Hs, b)Surge

AD or AI ?

- The choice of the framework depends on extremal correlations of the pair $(\ell(\mathbf{Z}_{M,1}), \ell(\mathbf{Z}_{M,2}))$ (see Figure 11).

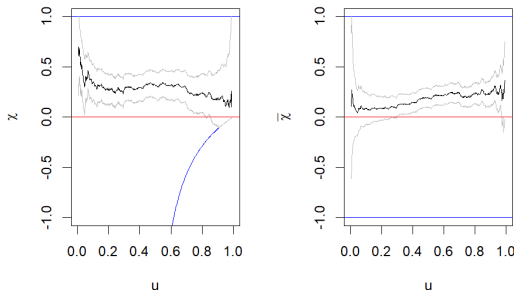


Figure 11: Extremal correlation coefficients for the pair S, H_s

AD or AI ?

- The choice of the framework depends on extremal correlations of the pair $(\ell(\mathbf{Z}_{M,1}), \ell(\mathbf{Z}_{M,2}))$ (see Figure 11). The vector of radial components exhibits asymptotic independence (AI) as $\chi \rightarrow 0$ and $\bar{\chi} \not\rightarrow 1$.

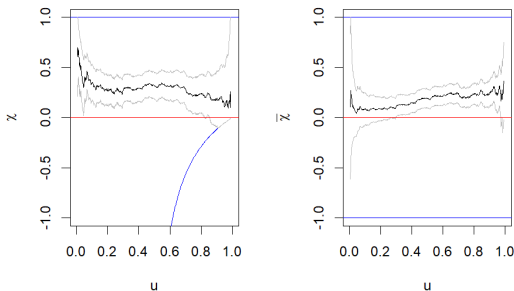


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Regular variations

- Following (Resnick 2002), regular variations remains valid in the AI case if the vector also exhibits hidden regular variations (HRV).

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- Two statistics can be used to verify HRV assumptions: Hillish and Pickandish statistics. The first one must converge in probability toward 1, the second toward 0 when the number of exceedances is low.

Framework used for radial components

- We can now determine the convenient framework in our case.

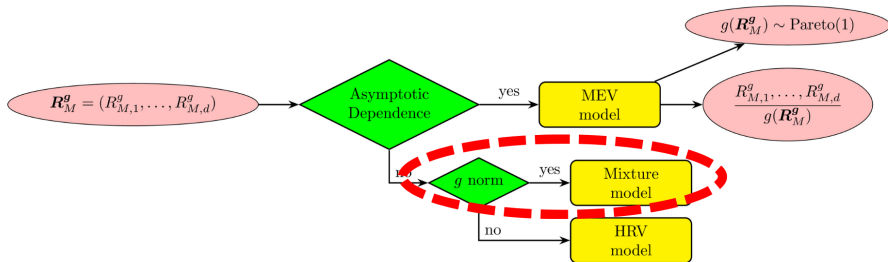


Figure 12: Model used in case of asymptotic independence

Framework used for radial components

- We can now determine the convenient framework in our case.
- Since the vector of radial components exhibits AI and g is a norm, we would use mixture models for the pair (S, H_s) (Figure 12).

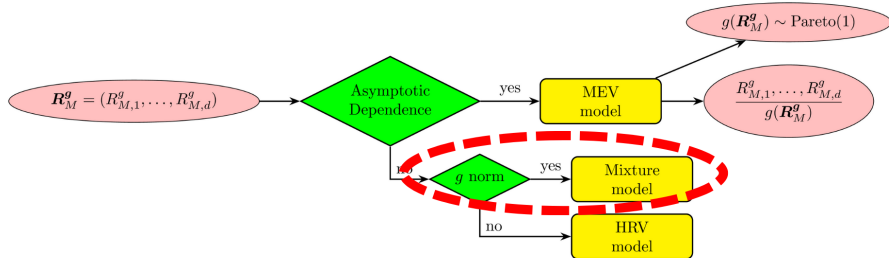


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Dependence of the radial component

- Table 2 presents results from model copula fitting. We find that the Gaussian model is the best model using the AIC and BIC criterion.
- Although the difference in AIC criterion is small between the best model and the other ones, the difference for the BIC criterion is reassuring. **However, we find much better results with the conditional model of (Heffernan and Tawn 2004)**

Number	Name	Log-L	par	par2	AIC	BIC
"1"	"Gaussian"	74.29	0.37	0	-144.59	-141.69
"2"	"t"	72.91	0.37	30	-141.83	-132.03
"14"	"Survival BB1"	71.95	0.31	1.12	-139.9	-130.1
"16"	"Survival BB7"	70.83	1.13	0.4	-137.65	-127.86
"10"	"BB8"	69.86	2.7	0.69	-135.73	-125.93

Table 2: Summary of model copula fitting

Simulations of $R_{M,1}^g, R_{M,2}^g$

- Figure 13 compares the simulation of radial components with extreme observations. The marginals are close in shape.
- In addition, the Kendall correlation coefficient of the simulated and observed vectors are close.

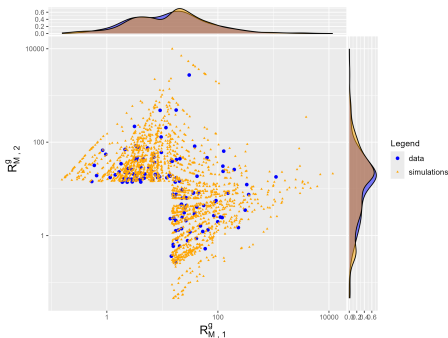


Figure 13: Simulation of radial components

- Now that we are able to approach the law of the radial vector, we approach the law of angular components.
- This modelling will enable us to simulate new extreme time series (see Figure 14).

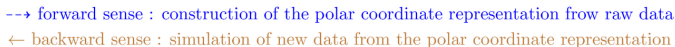


Figure 14: Modeling the law of angular component

Asymptotic dependencies in each variable

- As seen with the monivariate work with the surge time series, the pair $(\varepsilon_M^t, \varepsilon_M^s)$ exhibits asymptotic dependence for $t, s = 1, \dots, 37$ (Figure 15).

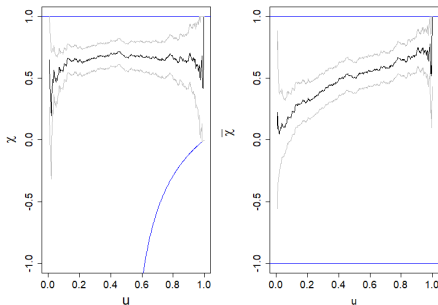


Figure 15: Extremal correlation coefficients for the surge at time $t = 1, s = 37$

Asymptotic dependencies for H_s

- The pair $(\varepsilon_M^t, \varepsilon_M^s)$ also exhibits asymptotic dependence for $t, s = 1, \dots, 37$ with H_s (Figure 16).

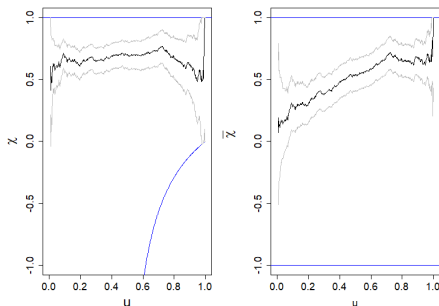


Figure 16: Extremal correlation coefficients for H_s at time $t = 1, s = 37$

Explanation of the variance

- Figure 17 describes the evolution of the unexplained inertia. We take 7 PCA components.

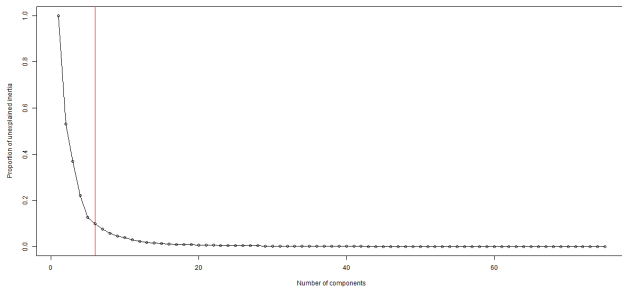


Figure 17: Evolution of unexplained inertia

- Figure 18 describes the simulation process resulting from the previous modeling steps.



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Consistency of the simulation

Visual comparison - - Surge

- We simulate this way 2000 multivariate extreme time series (against around 150 observed extreme time series).
- We represent a sample of simulated and observed extreme surge time series (Figure 19).

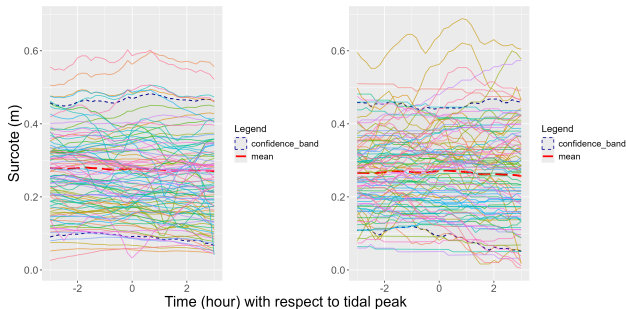


Figure 19: Simulated (left panel) and observed (right panel) extreme time series for the **surge**

Visual comparison - - Hs

- We simulate this way 2000 multivariate extreme time series (against around 150 observed extreme time series).
- We represent a sample of simulated and observed extreme Hs time series (Figure 20).

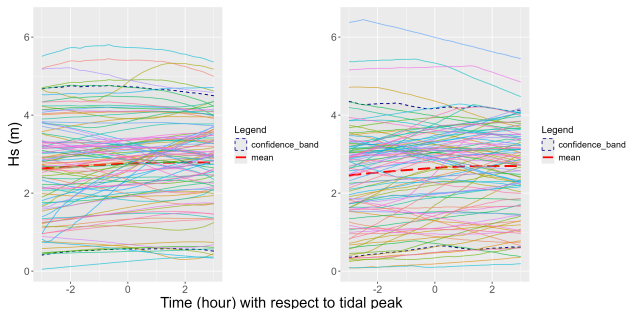


Figure 20: Simulated (left panel) and observed (right panel) extreme time series for **Hs**

Trends of the angular component -Hs

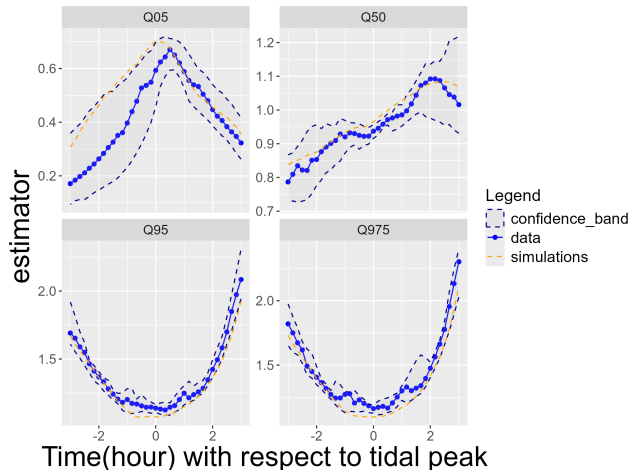


Figure 21: Evolution of several quantiles for $\Omega_{M,1}$

Trends of the angular component-Surge

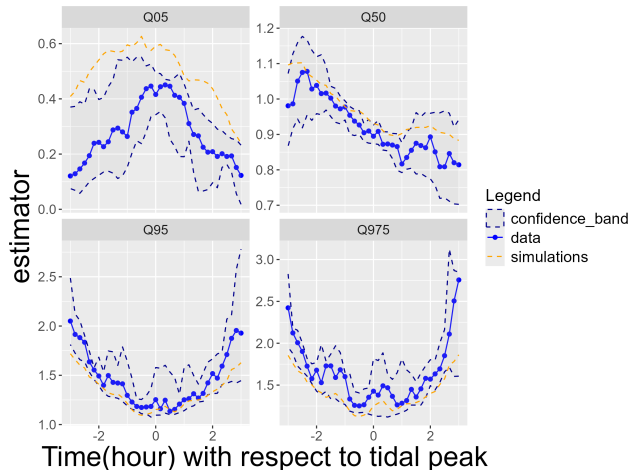


Figure 22: Evolution of several quantiles for $\Omega_{M,2}$

Distribution of the radial components in the original scale

- Figure 23 describes the distribution in the radial vector in extreme observations and simulations.

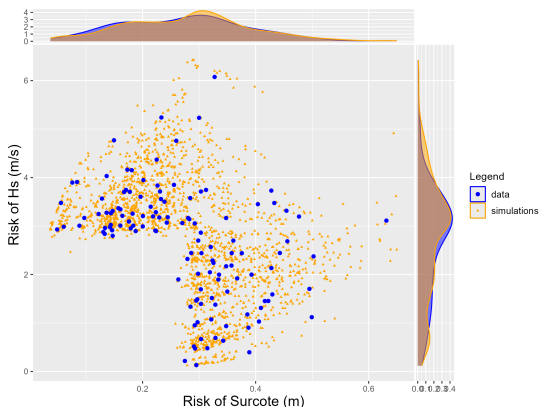


Figure 23: Distribution of risk functions

Trends-Hs

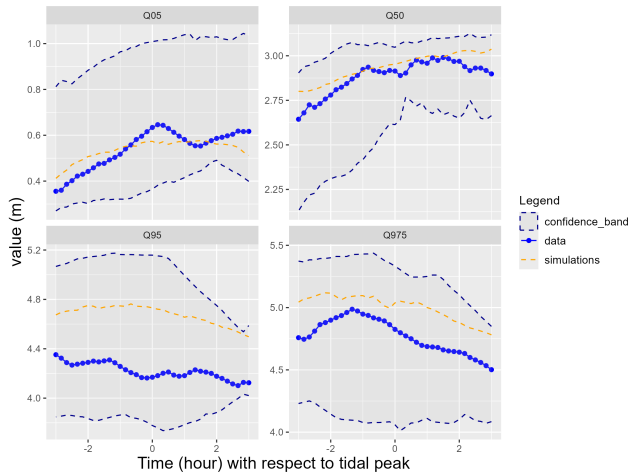


Figure 24: QQ results for Hs

Trends–Surge

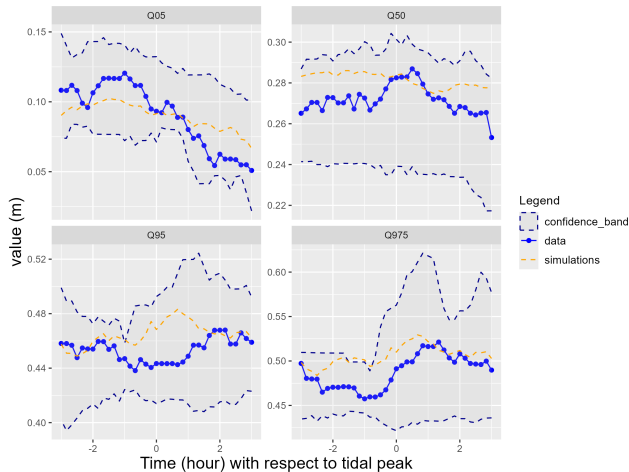


Figure 25: QQ results for the surge

PCA coordinates for surge time series

We can compare the PCA coordinates of observed and simulated extreme time series (Figure 26).

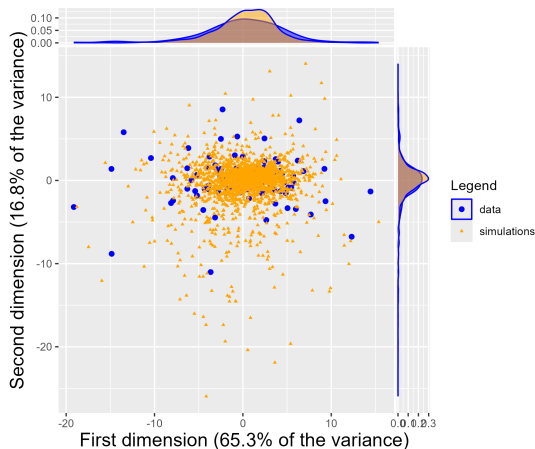


Figure 26: PCA coordinates for extreme surge time series

PCA coordinates for Hs time series

We can compare the PCA coordinates of observed and simulated extreme time series (Figure 27).

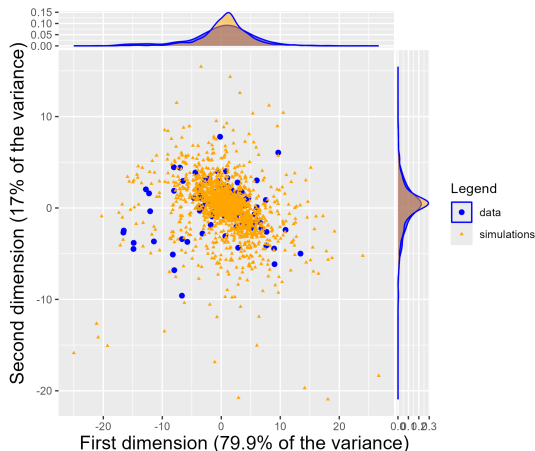


Figure 27: PCA coordinates for extreme Hs time series

Effect of the model

We have worse results with a EV model (Figure 28)

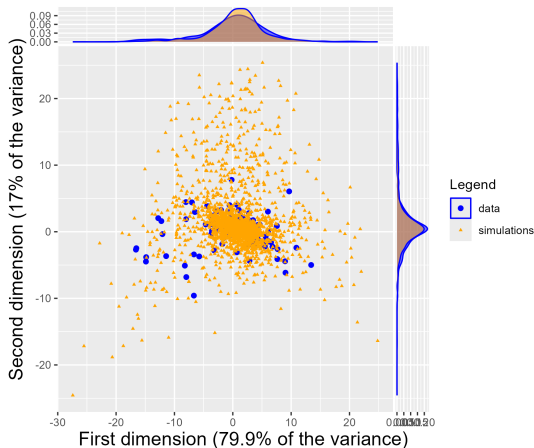


Figure 28: PCA coordinates for extreme Hs time series with logistic model

Comparing the correlations at each time t for angles

- Kendall correlation coefficient between $\Omega_{M,1}^t$ and $\Omega_{M,2}^t$ for each value of t for the extreme time series (Figure 29). Resampling methods to provide 95% confidence interval.

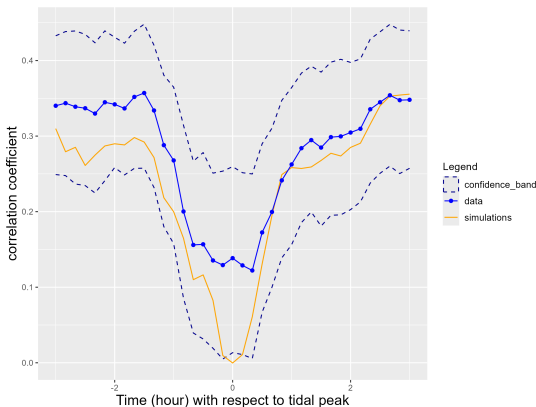


Figure 29: Correlation coefficients for each value of t

Comparing the correlations at each time t

- Kendall correlation coefficient between extremes of $\varepsilon_{M,1}^t$ and $\varepsilon_{M,2}^t$ for each value of t for the extreme time series (Figure 30). We retrieve here the difference seen for the radial components.

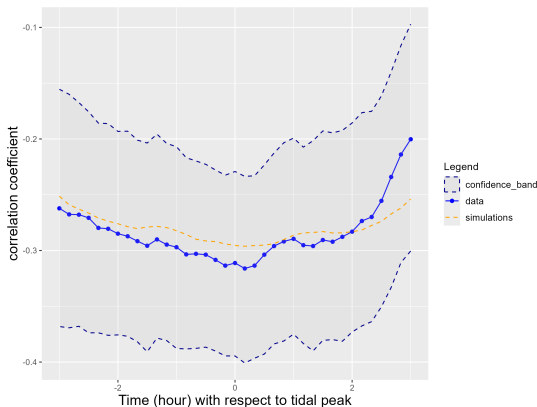


Figure 30: Correlation coefficients for each value of t

Distribution of $(\text{Surge}(t), H_s(s))$ for $(t = 13, s = 19)$

- Figure 31 shows what happens for (t, s) .

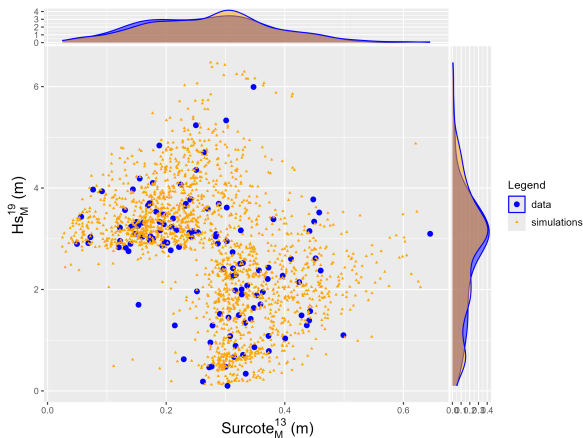


Figure 31: Distribution of $(\text{Surge}(t), H_s(s))$

Distribution of $(\text{Surge}(t), H_s(s))$ for $(t = 19, s = 37)$

- Figure 32 shows what happens for $t = 19, s = 37$.

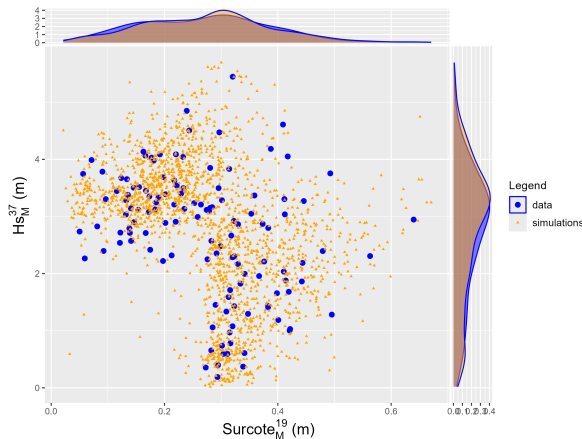


Figure 32: Distribution of $(\text{Surge}(t), Hs(s))$

Modelling of wind speed

Modeling of extreme wind speed time series

- We have now better results for the law of (H_s , Surge).
- These results come from the presence of asymptotic dependencies between at the different times for both H_s and Surge.
- If we want to use the trivariate case with U , we would need first to better approach the angular component of U .

Extremal dependencies of U time series

- The pair $(\varepsilon_M^t, \varepsilon_M^{t+h})$ exhibits asymptotic dependencies for some value of h (Figure 33)

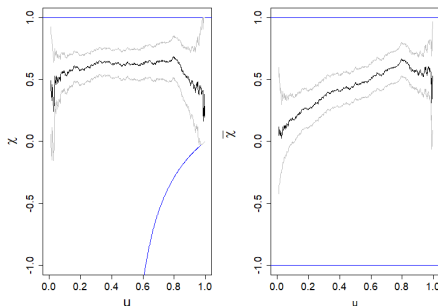


Figure 33: Extremal correlation coefficients for **wind speed** at time $t = 1, s = 19$

Extremal dependencies of U time series

- However, this dependence disappears when h is large (Figure 34). As noticed in (Rohrbeck and Cooley 2023), we cannot use the same method to model U 's angular component.

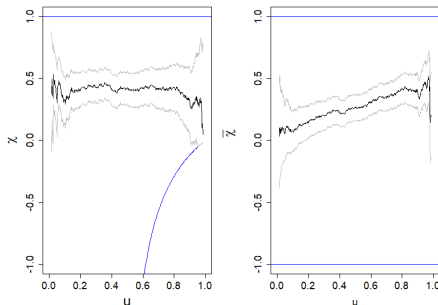


Figure 34: Extremal correlation coefficients for **wind speed** at time $t = 1, s = 37$

Summary of methods in the partial AD case

Deal with wind speed (Table 3).

Method	Summary	+	-
Hidden mixture model	Apply MEV models to a subvector	Flexible	Scalable ?
Spherical clustering	Identify clusters in the distribution of Ω_M	Use dimension reduction	Simulation not handled n_{ext} is small
Sparse PCA	Use only few active variables	Prevent putting mass in forbidden zones	Sufficient ?

Table 3: summary of partial AD models

Objectives

- Consider the several scenarios.
- It must have a link a common element between scenarios/ not enough observations to have independent modelling.

Proposition

- (Latent model) $y = x(z, W) + \varepsilon \in \mathbb{R}^d$ with $z \in \mathbb{R}^q$ latent variable, $\varepsilon \sim \mathcal{N}(0, \Sigma_\varepsilon)$. In our case, $d = 37$ and q is the number of PCA axes (Tipping 1998).

Proposition

- (Latent model) $y = x(z, W) + \varepsilon \in \mathbb{R}^d$ with $z \in \mathbb{R}^q$ latent variable, $\varepsilon \sim \mathcal{N}(0, \Sigma_\varepsilon)$. In our case, $d = 37$ and q is the number of PCA axes (Tipping 1998).
- (Factor analysis) $x(z, W) = Wz$ with $W \in \mathcal{M}_{d \times q}$ and $z \sim \mathcal{N}(0, \mathbb{I})$.

Proposition

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- (Factor analysis) $x(z, W) = Wz$ with $W \in \mathcal{M}_{d \times q}$ and $z \sim \mathcal{N}(0, \mathbb{I})$.
- (MPPCA) We may introduce a mixture distribution with M classes. Assuming y is generated from the k^{th} group, we have $y = W_k z + \varepsilon_k$.

Proposition

- (Latent model) $y = x(z, W) + \varepsilon \in \mathbb{R}^d$ with $z \in \mathbb{R}^q$ latent variable, $\varepsilon \sim \mathcal{N}(0, \Sigma_\varepsilon)$. In our case, $d = 37$ and q is the number of PCA axes (Tipping 1998).
- (Factor analysis) $x(z, W) = Wz$ with $W \in \mathcal{M}_{d \times q}$ and $z \sim \mathcal{N}(0, \mathbb{I})$.
- (MPPCA) We may introduce a mixture distribution with M classes. Assuming y is generated from the k^{th} group, we have $y = W_k z + \varepsilon_k$.
- W_k is obtained through the eigenvalue decomposition of the covariance matrix of the k^{th} group.

Simplified case

- If we use the MPPCA framework, we would apply a decomposition per cluster of observations.

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- Gaussian noise $\varepsilon_k \sim \mathcal{N}(0, \Psi_k)$ with a variance depending on the cluster.

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- If we use the MPPCA framework, we would apply a decomposition per cluster of observations.
- Gaussian noise $\varepsilon_k \sim \mathcal{N}(0, \Psi_k)$ with a variance depending on the cluster.
- Following parsimonious gaussian mixtures (McNicholas and Murphy 2008), we would impose a condition on the matrix W_k or ε_k . For instance, we would impose $W_k = W$ or $\Psi = \Psi_k$. $W_k = W$ gives the same decomposition for each cluster.

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